

Md Abdullah Al Mazid

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Summary

Machine-learning researcher and software engineer with 3+ years of industry experience and PhD research in multimodal AI, computer vision, remote sensing, and physics-guided learning. Build end-to-end ML systems from model design and GPU training through evaluation, APIs, and production-oriented applications.

Technical Skills

AI/ML: PyTorch, TensorFlow/Keras, computer vision, multimodal learning, diffusion models, ViTs, cross-attention, time-series modeling **Programming:** Python, C++, C#, SQL, Swift/SwiftUI

Systems/Tools: Linux, CUDA/GPU computing, Docker, Git, FastAPI, Core ML, Apple Vision, .NET, Microsoft SQL Server, Django, REST APIs

Experience

Research Assistant, AI/ML, High Performance Database Research Center, FIU Aug. 2022 – Present

- Build deep-learning systems for satellite imagery, atmospheric modeling, environmental forecasting, and multimodal data analysis using large GPU servers and reproducible training/evaluation pipelines.
- Develop models spanning optical–SAR fusion, diffusion-based restoration, Vision Transformers, cross-attention, physics-guided learning, and irregular spatiotemporal forecasting; contribute to NSF-funded research and multi-agency environmental projects.

Teaching Assistant, Florida International University Fall 2024, 2025, 2026

- Supported **Computer Systems Fundamentals** and **Principles of Database Management Systems** through lecture support, office hours, student discussions, assessment, and instructor-directed course design/review.

Senior Software Engineer, Enosis Solutions – Dhaka, Bangladesh June 2019 – Aug. 2022

- Developed and maintained Denticon, a large-scale cloud dental SaaS platform for Planet DDS, across backend features, legacy modernization, database development, and performance optimization.
- Led a development sub-team; optimized SQL queries and improved software quality, maintainability, and production workflows.

Software Engineer Intern, mPower Social Enterprises – Dhaka, Bangladesh May 2019 – June 2019

- Developed Python/Django software components for social-impact applications.

Selected AI/ML Projects

MSCloudr – Multimodal Cloud Removal with Sentinel-2 + Sentinel-1 SAR

- Developing a SAR-guided diffusion-bridge model for recovering cloud-obscured multispectral imagery from paired Sentinel-1 and Sentinel-2 observations.
- Designed mean-reverting bridge dynamics, reliability-aware conditioning, cross-modal alignment, and risk-aware trajectory variants to study stable and physically consistent multimodal reconstruction.

pKANrtm – Physics-Guided Atmospheric-Correction Surrogate

- Built a multi-fidelity Kolmogorov-Arnold Network using low-cost 6S guidance to emulate high-fidelity libRadtran coefficients for Sentinel-2 atmospheric correction; published in *Remote Sensing* (2026) and disclosed to FIU as invention D2026-0057 (accepted for review).

WheelBuddy – iOS AI Driving Assistant ShellHacks 2026

- Co-developed a SwiftUI app combining time-aware route crime-risk estimates with on-device vehicle/pedestrian, driver-attention, drowsiness, and lane perception using Core ML/Apple Vision; trained custom models and served predictions with FastAPI.

MSCloudCAM – Multispectral Cloud Segmentation

- Developed a ViT-based multi-sensor model with multi-scale context and cross-attention for four-class cloud segmentation on Sentinel-2 CloudSEN12 and Landsat-8 L8Biome; published at IEEE CAI 2026.

Selected Research Outputs

Remote Sensing 2026: pKANrtm | **IEEE CAI 2026:** MSCloudCAM | **IEEE SoutheastCon 2025:** ODE+BiLSTM water-quality forecasting | **WACV 2027:** Object-Grounded Text Priors for Saliency Ranking Detection

Education

Florida International University – PhD, Computer Science (Expected Spring 2027); MS, Computer Science (2024) |

Khulna University of Engineering and Technology – BS, Computer Science (2019)

Additional

Hackathons: ShellHacks (2022–2026) | **Competitive Programming:** ICPC Dhaka Regional Final (2015); National Collegiate Programming Contest (2016)